

Efficient Moral Judgment: Building a Lightweight Classifier for Ethical Decision-Making

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Abstract

Understanding and classifying moral norms is crucial for developing AI systems that make socially responsible decisions and interact appropriately in human-centered environments. In this work, we investigate methods for training lightweight models capable of moral judgment, leveraging the [Moral Stories](#) dataset [1]—a structured, crowd-sourced collection of narratives featuring both normative and divergent actions.

To balance predictive performance with computational efficiency, we employ parameter-efficient fine-tuning (PEFT) techniques such as Low-Rank Adaptation (LoRA), along with layer pruning and knowledge distillation. Our approach yields a compact classifier that retains strong moral reasoning capabilities while significantly reducing computational cost. We show that one of our models achieves 84% reduction in training, 85% in inference time and 77% in model size with <3% loss in classification accuracy, making it well-suited for deployment in real-world, resource-constrained settings.

1 Introduction

As AI systems increasingly influence real-world decision-making—from content moderation and autonomous agents to legal and healthcare applications—ensuring that these systems can understand and align with human moral norms is more important than ever. The ability to distinguish between morally acceptable and unacceptable behavior is fundamental to creating trustworthy, value-aligned AI. However, moral reasoning is inherently complex, context-sensitive, and shaped by cultural and societal

factors, making it difficult to formalize with traditional rule-based approaches.

Recent advancements in NLP, particularly with Transformer-based models, have enabled machines to interpret and classify nuanced textual data. While these models excel at tasks like sentiment analysis and fact-based classification, applying them to moral norm classification introduces unique challenges: moral judgments are often implicit and require contextual understanding beyond surface-level semantics. The Moral Stories dataset offers a promising foundation for training models to infer ethical alignment. However, fine-tuning large Transformer models for this task is computationally intensive, demanding significant resources in terms of memory, time, and hardware—resources that are not always accessible to researchers or practitioners.

This project addresses that gap by exploring resource-efficient techniques for training moral norm classifiers. Specifically, we apply layer pruning methods then healed by LoRA to reduce model size without compromising accuracy. Additionally, we investigate model distillation—using a compact, pre-distilled model like DistilBERT—to further enhance efficiency during both training and inference.

Our contribution is a comparative study of these strategies for building a lightweight yet effective moral classification model. We aim to demonstrate that moral reasoning capabilities can be retained even under tight computational constraints. By doing so, this research advances the development of scalable, ethics-aware AI systems and contributes practical insights to the intersection of NLP and AI ethics.

2 Background

Moral reasoning in AI requires more than surface-level classification—it demands an understanding of context, intent, and social norms. As highlighted in [Situated Reasoning about Norms, Intents, Actions, and Their Consequences](#) [2], ethical judgments depend not only on actions but also on the intentions behind them and the norms that govern specific situations. This underscores the need for models that can reason for morality in a situated, context-aware manner. The Moral Stories dataset directly supports this need by providing structured, crowd-sourced narratives that contrast normative and divergent outcomes, enabling models to learn from realistic moral scenarios.

However, the nuanced nature of moral judgment introduces challenges that go beyond simple classification. [Thinking Like a Skeptic: Defeasible Inference in Natural Language](#) emphasizes that human reasoning is often defeasible—subject to revision as new information becomes available [3]. This insight is especially relevant for moral classification, where judgments are rarely absolute and often depend on context that may be implicit or incomplete. Our work draws on this idea by training models to make flexible, context-sensitive ethical assessments based on narrative data.

While Transformer-based models have shown great promise for understanding complex language patterns, their computational cost remains a significant barrier to practical deployment. [The Unreasonable Ineffectiveness of the Deeper Layers](#) provides evidence that the deepest layers of large language models contribute minimally to downstream performance, suggesting that these layers can be pruned with little loss in accuracy [4]. We leverage this finding through layer pruning to reduce inference time and memory usage in our moral classification model.

To further enhance efficiency, our work incorporates PEFT strategies. [Exploring Versatile Generative Language Model via Parameter-Efficient Transfer Learning](#) demonstrates that training only a small fraction of parameters—around 2–3%—is sufficient for strong performance across tasks [5]. Inspired by this, we apply PEFT techniques such as LoRA to fine-tune moral reasoning models in a resource-conscious

manner, ensuring they remain lightweight while retaining high performance.

We also explored model distillation, a technique first introduced by Hinton et al. (2015), in which a smaller student model is trained to mimic the output distribution of a larger teacher model. This approach has shown promise in compressing large language models while retaining much of their performance [6]. Together, these insights provide the foundation for our approach: building a computationally efficient yet context-aware classifier for moral norms by combining narrative-driven training data with targeted model compression and adaptation techniques.

3 Methods

Our goal is to develop a computationally efficient yet accurate classifier capable of determining whether an action aligns with moral norms. To achieve this, we leverage the Moral Stories dataset and apply a combination of PEFT, layer pruning, and model distillation techniques. We also explored the combination of these techniques. The following subsections detail our methodology:

3.1 Baseline Model

As our primary baseline, we fine-tune RoBERTa-large, a robust Transformer-based language model known for its strong performance across a wide range of natural language understanding tasks. RoBERTa-large serves as a high-capacity benchmark against which we compare our efficiency-optimized models. The model is fine-tuned on the Moral Stories dataset using full parameter updates, allowing it to learn the complex moral reasoning patterns embedded in the narratives. During training we used the following hyper-parameters: learning rate of $2e-5$, weight decay of 0.05, batch size of 16, and maximum sequence length of 128.

In addition to RoBERTa-large, we also evaluated several other pre-trained Transformer architectures, including BERT-base, BERT-large, RoBERTa-base and DistilBERT. The results of these comparisons, including their relative performance and resource requirements, are provided in the Appendix A-2.

3.2 LoRA PEFT

To reduce training costs, we adopt LoRA, a PEFT method that injects trainable rank-decomposed matrices into the attention layers of the model while keeping the original pre-trained weights frozen. This enables us to achieve competitive performance while tuning only a small percentage of the model's parameters. We experiment with different rank values and learning rates to optimize the trade-off between efficiency and accuracy.

3.3 Layer Pruning

Motivated by findings from Gromov et al., we perform layer pruning to remove the deepest layers of the RoBERTa-large and RoBERTa-base model, which contribute least to task-specific performance. We prune between 2 to 18 layers from the 24-layer RoBERTa-large and 2 to 6 layers from RoBERTa-base and evaluate performance drop against improvements in model size and inference speed. This process is guided by validation performance and informed by ablation analysis.

3.4 Distillation of Models

In the context of transformer-based language models, distillation has become a widely adopted strategy to reduce the size and inference cost of models like BERT, without incurring a significant loss in performance. Several variants have emerged, including DistilBERT [7], which leverages a combination of loss functions—matching hidden states, attention scores, and logits—to train a compressed version of BERT that retains over 95% of its performance on downstream tasks while being 40% smaller and 60% faster. This process not only facilitates deployment in resource-constrained environments but also serves as a foundation for scalable experimentation in transfer learning and task-specific adaptation.

In this work, we explore the feasibility of building a fine-tuned model for classifying whether a given action in the Moral Stories dataset is moral or not. We leverage transfer learning with a distilled model that retains approximately 95% of the accuracy of larger models such as BERT-large or RoBERTa-large, while significantly reducing model size and inference cost.

In our experiments, we distill knowledge from a fine-tuned RoBERTa-large, RoBERTa-base and

DistilBERT teacher into progressively smaller student models by reducing the number of transformer layers. We assumed that pretraining the teacher model would give us an advantage for the student model, however no such advantage was observed in a few post work we conducted. We constructed 9 student models for RoBERTa-large, 3 for RoBERTa-base and 2 for DistilBERT varying in size from 22 layers down to just 1 layer, in decrements of 2 layers. For each model, we select a subset of layers from the original teacher to retain layers evenly spaced across the depth — this means outer layers (closer to the input and output) are preserved while center layers are more likely to be dropped. The student reuses the teacher's embedding and classifier weights, and during training, the first 6 layers are kept frozen from training while others are fine-tuned. Knowledge distillation is supervised using a combination of cross-entropy with gold labels and KL divergence between student and teacher logits (temperature=1.5, $\alpha=0.5$), optimized with AdamW at a learning rate of $2e-5$ for 4 epochs. We evaluate each student model in terms of validation accuracy and inference time (averaged over 500 samples) to quantify the trade-offs between model size, accuracy, and efficiency.

3.5 Evaluation Metrics

- Measure **accuracy** to assess classification performance. Accuracy is picked as the evaluation metrics because the dataset is balanced, and there is no material difference between accuracy and F1, so accuracy alone is sufficient.
- **Training time, inference time, and model size** for computational efficiency

4 Results and Discussion

To ensure a fair comparison across methods and unless otherwise stated, training time and accuracy results are reported from the first epoch, with hyperparameters selected to optimize each model's accuracy performance.

4.1 Baseline and partial fine-tuning

Among the various fine-tuned Transformer models evaluated, RoBERTa-large consistently achieved the highest accuracy and was consequently selected as the baseline model (see Appendix A-2). Given that the central objective

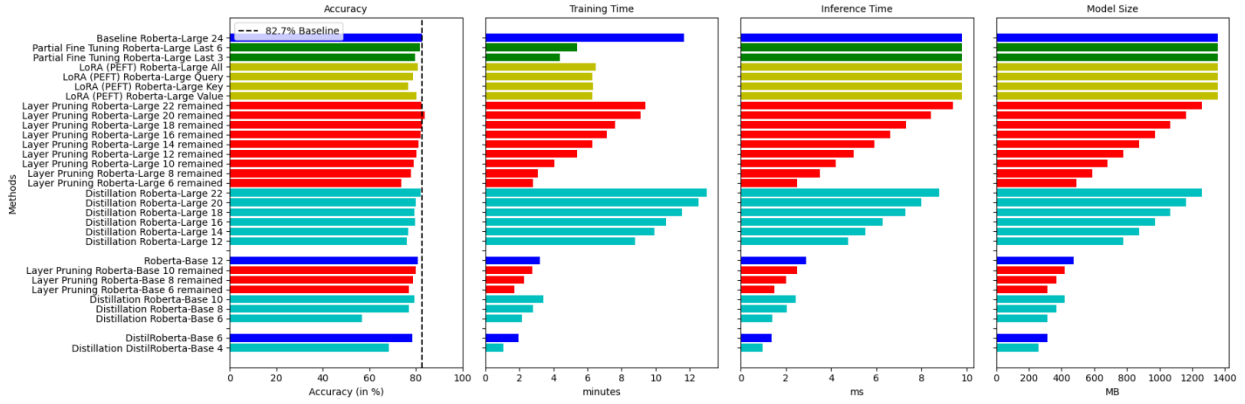


Fig 1: Performance of various models built using pruning and distillation (Left) Accuracy of the model vs. the model size of that model (Center) Accuracy of the model vs. Inference time for that model (Right) Accuracy of the models vs. the training time for 1 epoch of that model.

of this research is to maintain strong classification performance while reducing computational cost and training time, we explored partial fine-tuning by updating only selected layers of the model. Fig 1 summarizes the performance of RoBERTa-large when fine-tuned on different subsets of its layers. Fine-tuning only the last 6 layers, as opposed to all 24 layers, resulted in over 50% reduction in training time with only a marginal drop in accuracy (from 82.7% to 81.6%). In contrast, fine-tuning just the last 3 layers further decreased training time but introduced a more pronounced decline in accuracy (to 79.5%). These findings highlight that significant computational savings can be achieved with minimal impact on model performance by selectively fine-tuning upper layers of RoBERTa-large.

4.2 PEFT via LoRA

To further improve training efficiency while maintaining strong performance, we applied LoRA to the RoBERTa-large model. LoRA enables parameter-efficient fine-tuning by introducing trainable low-rank matrices into specific components of the attention mechanism—namely, the query, key, and value projections—while keeping the rest of the model weights frozen.

Fig. 1 (also available in Appendix A-4) summarizes the results of various LoRA configurations. Fine-tuning all attention components with LoRA achieved the highest accuracy at 80.9%, while still reducing training time by 44% compared to full fine-tuning. This demonstrates that LoRA can preserve most of the model’s predictive power with significantly lower training overhead.

Among the component-wise adaptations, applying LoRA only to the value projections achieved the second-best accuracy (80.1%) with comparable training time. In contrast, isolating LoRA to only the query or key projections led to more noticeable drops in accuracy (78.9% and 76.8%, respectively), suggesting that value projections play a more critical role in moral classification tasks within the RoBERTa architecture. Notably, selectively applying LoRA to specific components of the attention mechanism did not further reduce training time and came with trade-offs in accuracy.

These results support the effectiveness of LoRA as a lightweight and flexible adaptation method for moral classification, allowing for considerable reductions in training cost while retaining much of the baseline model’s performance. When compared with partial fine-tuning, however, LoRA shows more degradation in accuracy.

4.3 Layer Pruning

Training efficiency is one of the primary advantages of partial fine-tuning and PEFT methods, but they do not save on inference time. To explore structural model efficiency, we performed a series of layer pruning experiments on RoBERTa-large, progressively reducing the number of layers and then fine-tuned the model. The aim was to examine the trade-offs between model depth, accuracy, and computational cost.

As shown in Fig. 1, pruning RoBERTa-large from 24 to 18 layers resulted in over 40% reduction in training time (from 10:55 min to 7:38 min) and 25% reduction in inference time (from 9.8 ms to 7.3 ms), with no loss in accuracy

(remaining at 82.7%). Interestingly, the highest accuracy (83.9%) was achieved at 20 layers, though this may be due to random variation rather than a consistent effect. However, pruning below 14 layers led to more pronounced accuracy declines, with performance dropping to 73.7% at just 6 layers—indicating that overly pruned models lack the representational capacity needed for nuanced moral reasoning.

model, which may not retain the necessary richness for effective teaching.

Overall, RoBERTa-large achieves an accuracy of 82.7% with its 24-layer architecture, requiring 1.4 GB of memory and taking around 10 ms for inference on a single test case. In comparison, a 6-layer DistilRoBERTa-base reaches 80% accuracy, with a 77% reduction in memory usage and 85% reduction in inference time. Our results provide a

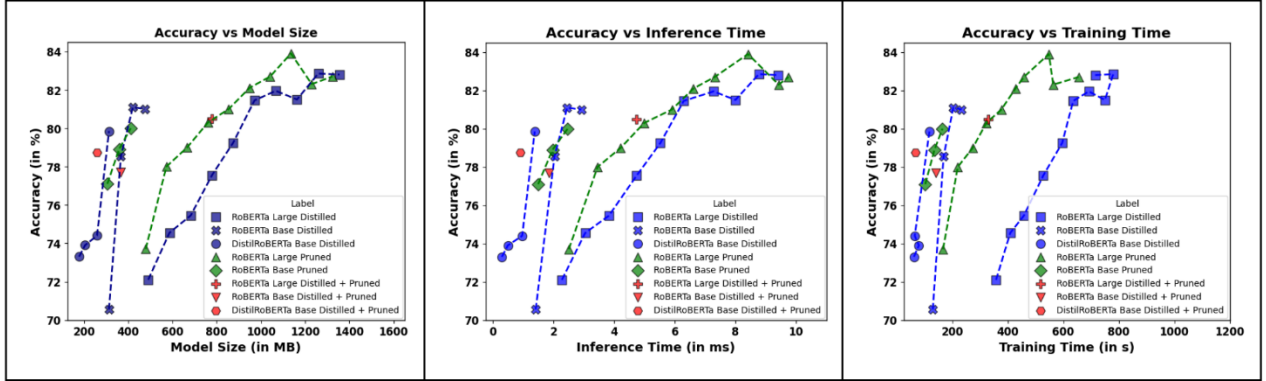


Fig. 2 Accuracy, training time, and inference time by various methods for 1 epoch.

Notably, a pruned 12-layer RoBERTa-large model performed worse than the full 12-layer RoBERTa-base model. This observation led us to repeat the pruning experiments on RoBERTa-base, where we found it achieved better performance across accuracy, training time, and inference latency when compared at equivalent depths.

4.4 Model Distillation

Three models—RoBERTa-large (24 layers), RoBERTa-base (12 layers), and DistilBERT (6 layers)—were distilled into smaller student models. Fig 1 illustrates the performance as the larger models are distilled into progressively smaller versions. We observe a marginal decrease in accuracy as the number of layers in the student model is reduced. As expected, both the inference time and model size decrease with smaller models. While the accuracy reduction is gradual as we transition from larger to smaller layers, the rate of accuracy loss accelerates the further we deviate from the original teacher model. Additionally, we observe a significant drop in accuracy when further distilling DistilBERT, likely because distilling from an already compressed model amplifies performance degradation due to compounding approximation errors. This happens as we train the student on a simplified, less powerful version of the original

comprehensive performance comparison across the parameter space between RoBERTa-large and DistilRoBERTa-base, offering insights into the trade-offs between accuracy, memory requirements, and inference time.

When distilling RoBERTa-large and RoBERTa-base models into smaller counterparts, we observed that the resulting models underperformed compared to pre-trained models of similar size. For instance, a RoBERTa-large model distilled to 12 layers achieved lower accuracy than the RoBERTa-base model, which is natively 12 layers. Similarly, the distilled RoBERTa-base model (12 layers) performed worse than the pre-trained DistilBERT model (6 layers). In both cases, the distilled models lagged behind their pre-trained counterparts in accuracy and other performance metrics.

We hypothesize that this performance gap could be mitigated through more careful tuning of hyperparameters such as the temperature (T), which controls the softness of the teacher's output distribution, and the distillation loss weight (α). However, this analysis was beyond the scope of the current work.

5 Conclusion

In summary, our experiments demonstrate that substantial gains in computational

efficiency can be achieved through a combination of partial fine-tuning, parameter-efficient methods like LoRA, structural compression via layer pruning, and knowledge distillation—each offering distinct trade-offs. Partial fine-tuning and LoRA effectively reduce training cost with minimal to moderate impact on accuracy, while moderate amount of pruning and distillation reduce inference latency and model size while maintaining accuracy. It is worthy to note that aggressive pruning or distillation, especially from already compressed models, resulted in noticeable performance degradation.

Perhaps most notably, our findings show that simply fine-tuning a strong lightweight model—such as RoBERTa-base or DistilBERT—can outperform distilled or pruned variants of larger models at comparable or smaller sizes. This highlights the importance of selecting well-pretrained architectures over relying solely on compression techniques. Taken together, these results underscore that the path to efficient moral reasoning models lies not only in model compression but also in smart model selection. Our work provides a foundation for developing lightweight, accurate models for moral classification and invites further research into hybrid approaches and optimization of distillation strategies.

6. Future Work

To save time, we reported validation results after only one training epoch; in future experiments, we plan to increase this up to three epochs for more stable evaluation.

One key observation is the significant drop in accuracy observed in distilled models as the number of layers is reduced further from the original architecture. We hypothesize that this degradation could be mitigated through careful hyperparameter tuning,

particularly of the distillation temperature (T) and the loss weighting factor (α).

In Fig 2, we report experimental results of further model compression using a combination of model pruning and distillation. However, these configurations were derived from intuitive combinations of parameters rather than the result of a principled search or full ablation study. These initial results are promising, but more extensive experimentation is needed to properly evaluate the interaction effects of pruning and distillation strategies.

We also plan to investigate further compression techniques, such as reduced-precision training and inference using float16, which could lead to additional gains in model size and inference speed without substantial loss in accuracy.

Finally, the distillation process itself could be improved by incorporating lessons from the development of DistilBERT (Sanh et al., 2019), which carefully balances layer selection, intermediate supervision, and loss design to produce a highly efficient student model. Adapting such insights to our domain-specific task may yield better-performing lightweight models.

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Contributions

Hitesh Arjun Basantani and Alice Robinson collaborated closely on all aspects of this project. Their individual contributions are outlined below:

Hitesh Arjun Basantani was responsible for the design and execution of the distillation experiments, including student-teacher training pipelines and layer selection strategies. He also contributed to editing and synthesizing the final report.

Alice Robinson led the implementation of model pruning and LoRA-based PEFT experiments. She also contributed significantly to the writing of the Methods, Results, and Conclusion sections.

Both authors collaboratively conducted the literature review and framed the background context. They jointly wrote the Abstract, Introduction, and Background sections, and co-developed the visualizations. Additionally, they participated equally in discussions around methodology, model evaluation, and final manuscript preparation.

A Appendices

A-1 Data

Each story in the *Moral Stories* dataset is associated with a unique norm selected from the recently published SOCIAL-CHEM-101 (SC-101) dataset (Forbes et al., 2020), provides a structured collection of narratives that depict morally normative and divergent behaviors, along with their consequences. Norms were crowd-sourced from contributors residing predominantly in the US and therefore, reflect values endemic to

the US society. There are 12k short, structured stories, and each story consists of seven sentences belonging to the following categories:

Norm: A guideline for social conduct generally observed by most people in everyday situations.

Situation: Setting of the story that introduces story participants and describes their environment.

Intention: Reasonable goal that one of the story participants (the actor), wants to fulfill.

Normative action: An action by the actor that fulfills the intention and observes the norm.

Normative consequence: Possible effect of normative action on the actor’s environment.

Divergent action: An action by the actor that fulfills the intention and diverges from the norm.

Divergent consequence: Possible effect of the divergent action on the actor’s environment.

Example data points below:

```
{
  'ID': '3B1NLC6UG0U92AG04KLH7KSR3IGGPA1',
  'norm': "It's discouraged to play with
someone's dog without asking first.",
  'situation': 'Joshua sees his neighbor
walking down the street with his dog.',
  'intention': 'Joshua wants to interact with
the dog.',
  'moral_action': 'Josh asks his neighbor if
he could play with his dog.',
  'moral_consequence': 'The neighbor consents
and Josh is happy as he plays with the dog.',
  'immoral_action': 'not specified',
  'immoral_consequence': 'not specified',
  'label': 1
}

{
  'ID': '3P4RDNWND64RCQJXZVGV0JVAE6IJ90',
  'norm': "It's wrong to date underage
people.",
  'situation': 'Jeff was scrolling through
potential dates on a dating app that he had
just downloaded.',
  'intention': 'Jeff wants to get a date.',
  'moral_action': 'not specified',
  'moral_consequence': 'not specified',
  'immoral_action': 'Jeff asks a sixteen year
old that he finds on the app if she wants to
meet him.',
  'immoral_consequence': 'Jeff is arrested
and sent to jail for dating a minor.',
  'label': 0
}
```

The length distribution of sentences from each category is normally distributed, with an average of 44 words in norm, 81 words in both normative and divergent actions, 76 words in moral consequence and 77 words in divergent consequences.

This dataset offers a rich resource for training models to recognize moral norms by presenting

situations with clear ethical distinctions. The structured nature of the dataset allows models to learn how different actions lead to different consequences, making it particularly useful for moral norm classification tasks.

More detailed results can be found below.

A-2 Transformer Model Comparison

We evaluated the performance of several Transformer models using the same hyperparameters and without freezing any layers. As shown in Table 1, RoBERTa-large consistently outperformed the others, achieving the highest accuracy across different levels of grounding.

Model	Action	+Norm	+Conseq.
roberta-base	81.6%	85.4%	98.9%
roberta-large	82.7%	92.3%	99.1%
twitter-roberta-base-sentiment	81.0%	83.6%	98.3%
bert-base-case	79.1%	79.2%	97.5%
bert-large-case	80.5%	82.7%	98.0%
deberta-v3-base	82.7%	92.3%	98.6%

Table 1: Accuracy results for Transformer models at different grounding levels.

One possible reason for RoBERTa's strong performance is that it was trained on a larger dataset and uses dynamic masking training architecture. These aspects improve its contextual understanding, which is crucial when dealing with complex moral reasoning tasks that require nuanced interpretation of context, action, and consequences.

It is expected that accuracy improves with the increase in grounding. When consequence is included as part of the grounding, the model achieves near-perfect accuracy. However, this can be attributed to the worker's tendency to associate socially acceptable actions with positive consequences. Since the ultimate goal is to enable AI systems to classify morally acceptable behavior, the remainder of the research focuses on action-only grounding.

A-3 Detailed Partial fine-tuning results

Layers Trained	All 24	Last 6	Last 3
Accuracy	82.7%	81.6%	79.5%
Training Time	11:39	5:23	4:22

Inference Time	9.8ms	9.8ms	9.8ms
Model Size (mb)	1,356	1,356	1,356

Table 2: Accuracy, training time (using L4 GPU), inference time, and model size of RoBERTa-large model with different layers trained.

A-4 Detailed LoRA results

Table 3 presents the results of additional LoRA experiments, evaluating different hyperparameter settings and attention component configurations. The inference time and model size are not included because they are consistent with all LoRA setups (9.8ms and 1,356mb respectively)

Layers	r	alpha	Accuracy	Training Time
All	8	8	79.5%	7:06
All	8	16	78.9%	7:02
All	8	24	80.6%	7:02
All	8	32	80.9%	6:42
All	4	12	80.2%	7:06
All	2	6	78.7%	7:00
All	1	3	76.7%	6:55
Query	8	24	78.9%	6:54
Key	8	24	76.8%	6:54
Value	8	24	80.1%	6:50

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679 **Table 3:** LoRA performance by hyperparameter

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681 **A-5 Detailed Layer pruning results**

682 This section explores the trade-off between
683 accuracy and computational efficiency when
684 applying layer pruning techniques to both
685 RoBERTa-large and RoBERTa-base models.
686 Notably, no significant loss in accuracy is
687 observed until the model is pruned down to 14
688 layers, while significant savings of computational
689 resources are saved.

Base Model	Remaining Layer	Full Healing		LoRA Healing		Inference Time	Model Size (mb)
		Accuracy	Training Time	Accuracy	Training Time		
roberta-large	24	82.7%	10:55	80.6%	7:02	9.8ms	1,356
roberta-large	22	82.3%	9:24	79.1%	6:32	9.4ms	1,259
roberta-large	20	83.9%	9:08	77.9%	6:02	8.4ms	1,163
roberta-large	18	82.7%	7:38	77.0%	5:28	7.3ms	1,067
roberta-large	16	82.1%	7:08	76.8%	4:48	6.6ms	971
roberta-large	14	81.0%	6:17	75.8%	4:15	5.9ms	875
roberta-large	12	80.3%	5:23	73.2%	3:38	5.0ms	779
roberta-large	10	79.0%	4:34	73.3%	3:03	4.2ms	683
roberta-large	8	78.0%	3:40	72.8%	2:25	3.5ms	587
roberta-large	6	73.7%	2:47	62.5%	1:47	2.5ms	491
roberta-base	12	80.7%	3:12			2.9ms	475
roberta-base	10	80.0%	2:45			2.5ms	421
roberta-base	8	78.9%	2:16			2.0ms	367
roberta-base	6	77.1%	1:43			1.5ms	313

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693 **Table 4:** Accuracy, training time (using L4 GPU),
694 and inference time results when after layer pruning.

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A-6 Detailed Distillation results

Model	Layers	Distilled Layers	Training Time (sec)	Parameters (in million)	Accuracy (in %)	Inference Time (ms)	Model Size (MB)
RoBERTa Large	24	24	715	355.36	84.5	9.402	1355.6
RoBERTa Large	24	22	780	330.16	82.85	8.772	1259.5
RoBERTa Large	24	20	750	279.78	81.5	7.996	1163.39
RoBERTa Large	24	18	692	254.59	81.95	7.279	1067.29
RoBERTa Large	24	16	636	229.39	81.45	6.291	971.19
RoBERTa Large	24	14	596	204.20	79.25	5.52	875.09
RoBERTa Large	24	12	528	179.01	77.55	4.739	778.99
RoBERTa Large	24	10	458	153.82	75.45	3.821	682.89
RoBERTa Large	24	8	409	128.62	74.55	3.039	586.79
RoBERTa Large	24	6	358	103.43	72.1	2.272	490.68
RoBERTa Base	12	12	234	124.64	81	2.912	475.49
RoBERTa Base	12	10	204	110.47	81.1	2.445	421.42
RoBERTa Base	12	8	169	96.29	78.55	2.028	367.34
RoBERTa Base	12	6	130	82.11	70.55	1.406	313.26
DistilRoBERTa Base	6	6	118	82.11	79.85	1.375	313.26
DistilRoBERTa Base	6	4	65.00	67.94	74.4	0.96	259.19
DistilRoBERTa Base	6	2	80	53.76	73.9	0.489	205.11
DistilRoBERTa Base	6	1	62	46.68	73.3	0.29	178.07
RoBERTa Large Distilled + Pruned	24	16	330	204.20	80.5	4.281	778
RoBERTa Base Distilled + Pruned	12	8	140	96.29	77.7	1.851	367.34
DistilroBERTa Distilled + Pruned	6	5	68	60.85	78.75	0.62	259.19

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Table 6: Model Name, Model Layers, Distilled Layers, Training Time (in seconds), Parameters (in millions), Accuracy (in %), Inference Time (ms) and Model Size (in MB) after distilling large models to student models with smaller layers.